

1.INTRODUCTION

Kudumbashree is a poverty eradication and women empowerment initiative implemented by the Government of Kerala in 1998. It is considered one of the largest women-led networks in the world, focusing on socio-economic development and inclusion. Kudumbashree operates at the grassroots level through a three-tier structure, empowering women through neighborhood groups (NHGs), area development societies (ADS), and community development societies (CDS). This project focuses on a individual Kudumbashree Unit (NHG).

The Kudumbashree Unit Automation aims to support the management and day-to-day activities of individual Kudumbashree unit, streamlining record-keeping, member management, and financial transactions. This system will not only enhance operational efficiency but also contribute to the transparency and effectiveness of the units' functioning.

1.1 OBJECTIVE OF THE PROJECT

The objective of the Kudumbashree Unit Automation project is to develop an efficient and user-friendly system designed to streamline the operations of a Kudumbashree unit. This system aims to automate key tasks such as member management, attendance management, loan notification, reports storage and reducing the need for manual record-keeping. By enhancing accuracy in data management the system will improve the overall efficiency of unit operations. Ultimately, the project seeks to enhance the organizational effectiveness of Kudumbashree units, enabling them to better meet their socio-economic goals.

Simply, Kudumbashree Unit Automation is a website to digitalize the heavy load paper works of a Kudumbashree Unit. It is an integrated package which has multiple features to assist the users such as secretary and other members. It is basically looking to provide a computerized atmosphere for the Kudumbashree members, that they can make an easy way to complete their paper work.

2. SYSTEM ANALYSIS

System analysis is a step-by-step process used to identify and develop or acquire the software need to control the processing of specific application. System analysis is a continuing activity the stages of the systems development. System analysis is the process of gathering and interpreting facts, diagnosing problems and using the facts to improve the system. The outputs from the organization are traced through the various processing that the input phases through in the organization. This involves gathering information and using structured tools for analysis. A detailed study of this process must be made by various techniques like interviews; questionnaires etc.

2.1 EXISTING SYSTEM

The existing system is being operated in manually. The entire record keeping is manually. This method of operation is very time consuming i.e. for managing information about the unit leader and members details. This involves lot of manual effort; more personal and storage space is less required. The data may be lost with the passage of time. So, the problem we given to solve is to build an automated system which helps the administrator, unit leaders and the all members in the system.

DISADVANTAGES OF EXISTING SYSTEM

1. High Risk of Data Duplicity: The current system, being manual or semi-digital, increases the chances of data being duplicated, leading to errors and inconsistencies in financial and member-related records.
2. Tedious Updating Process: Any updates to records, such as membership changes or financial transactions, require manual intervention, which can be slow and prone to human error. This inefficiency makes maintaining accurate data difficult.
3. Large Number of Records: Managing a large volume of records without a proper digital system becomes cumbersome. This is especially true for units with many members, as handling paper records or outdated systems leads to inefficiencies and difficulties in retrieving data.

4. Lack of Systematic Organization and Security: Without a structured digital system, there is no consistent way to organize, store, or secure important information. Sensitive data could be lost, misplaced, or accessed without proper authorization.

5. Time-Consuming Processes: Most tasks, from record-keeping to report generation, are time-consuming under the current system. The manual nature of operations slows down overall productivity and delays decision-making processes.

2.2 PROPOSED SYSTEM

The Kudumbashree Unit Management System is a website designed to enhance the operational efficiency and effectiveness of Kudumbashree units. This system automates tasks such as member management, meeting scheduling, and program tracking, attendance tracking etc. It features a centralized database to store and manage data securely, a user-friendly interface for easy access and navigation by administrators, unit leaders, and members, and robust reporting tools for monitoring progress and evaluating outcomes.

ADVANTAGES OF PROPOSED SYSTEM

1. User-friendly interfaces: The system provides user-friendly interfaces with efficient registering and log in. Several validation routines are also provided which ensure error free addition of records.
2. Easier access to information: It is easier to access information from the available set of records. Searches can be easily done with the highest precision of accuracy.
3. Speed: The most important advantage of the computerized system is the speed with operations can be performed.
4. Avoidance of Redundancy: It ensures that records are redundant that is avoiding duplication of records. In this site the customers are unique by their log in details.
5. Consistency: The major problem that can be overcome is the problem of consistency. It ensures that changes made are available to all.
6. Security: It provides security by means of passwords.

7. Centralized Data Management: Provides a centralized database to store and manage all member information, and activity records, ensuring easy access and consistency.

2.3 SYSTEM REQUIREMENT SPECIFICATION

A software requirements specification (SRS) is a comprehensive description of the intended purpose and environment for software under development. The SRS fully describes what the software will do and how it will be expected to perform. An SRS minimizes the time and effort required by developers to achieve desired goals and also minimizes the development cost. A good SRS defines how an application will interact with system hardware, other programs and human users in a wide variety of realworlded situations.

Problem to be solved

The problem to be solved is the inefficiency and lack of security in the current Kudumbashree management system, which relies on manual processes.

Customer requirements

- The system should be fast.
- User-Friendly Interface
- Efficient Record Management
- Secure Data Storage
- Time Efficiency

What the developer needs to know?

- Must know the existing system and its drawbacks.
- Must know what will be needed in the proposed system.

Bussiness Requirements

The system should be feasible both to the developer and client. It should be effective and should be able to complete in time. Developer should be responsible for developing the system, install

the software and update the software whenever necessary, conducting any user training that may be needed for using the system.

User Requirements

The user requirement(s) specification is a document usually that specifies the requirements the user expects from software to be constructed in a software project.

Administrator has overall control in the system.

- Admin has overall control in the system.
- Provide secretary details.
- Faster processing.

Functional Requirements

Functional requirements define what a system is supposed to do. The system should perform the following functionalities.

- Login – Login of admin, secretary, members.
- Registration – Admin can register secretary with the system. Members register through a sign up form.
- Edit profile – Secretary and Members can edit and update their profile.
- Add Reports – Secretary can add meeting, program and attendance reports.
- Loan notification – Secretary shares the details of any available loan and members can apply if needed.
- View reports – Admin and Members can view reports of programs and meetings.
- View Attendance – Members can view their attendance.

- Logout – Secretary and Members can logout from panel.

2.3.1 HARDWARE SPECIFICATION

Processor	: Intel core i5
Speed	: 3.20GHz
System Bus	: 64 bits
Memory	: 8GB RAM
Hard disk	: 40GB or higher
Monitor	: 15.6-inch
Keyboard	: 104 keys
Pointing device	: Touchpad

2.3.2 SOFTWARE SPECIFICATION

Operating System	: Windows
Front End	: PHP, HTML, CSS
Scripting Language	: Java script
Back End	: SQL Server Web Server XAMPP 3.3.0
Browser	: Google Chrome

2.3.3 FRONT END

HTML

Hypertext Markup Language (HTML), the languages of the World Wide Web (WWW), allows users to produce Web pages that include text, graphics and pointer to other Web pages (Hyperlinks). HTML is not a programming language, but it is an application of ISO Standard 8879, SGML (Standard Generalized Markup Language), but specialized to hypertext and adapted to the Web. The idea behind Hypertext is that instead of reading text in rigid linear structure, we can easily jump from one point to another point. We can navigate through the information based on our interest and preference. A markup language is simply a series of elements, each delimited with special characters that define how text or other items enclosed within the elements should be displayed. Hyperlinks are underlined or emphasized works that load to other documents or some portions of the same document.

CSS

Cascading Style Sheets (CSSs) is a style sheet language used for describing the presentation of a document written in a mark-up language. Although most often used to set the visual style of web pages and user interfaces written in HTML and XHTML. The language can be applied to any XML document, including plain XML. SVG and XUL and is applicable to rendering in speech, or on other media.

Along with HTML and JavaScript, CSS is a cornerstone technology used by most websites to create visually engaging webpages, user interfaces for web applications, and user interfaces for many mobile applications.

CSS is designed primarily to enable the separation of document content from document presentation, including aspects such as the layout, colours, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification or presentation characteristics, enable multiple HTML pages to share formatting by specifying on all platforms except Windows. MySQL ships with no GUI tools to administer MySQL databases or manage data contained within the databases. Users may use the included command

line tools or install MySQL Workbench via a separate download. Many third party GUI tools are also available.

JAVA SCRIPT

JavaScript is a script-based programming language that was developed by Netscape Communication Corporation. JavaScript was originally called Live Script and renamed as JavaScript to indicate its relationship with Java. JavaScript supports the development of both client and server components of Web-based applications. On the client side, it can be used to write programs that are executed by a Web browser within the context of a Web page. On the server side, it can be used to write Web server programs that can process information submitted by a Web browser and then update the browser's display accordingly. Even though JavaScript supports both client and server Web programming, we prefer JavaScript at Clientside programming since most of the browsers supports it. JavaScript is almost as easy to learn as HTML, and JavaScript statements can be included in HTML documents by enclosing the statements between a pair of scripting tags, `<SCRIPTS>...</SCRIPT>`.

`<SCRIPT LANGUAGE = "JavaScript">`

JavaScript statements

`</SCRIPT>`

2.3.4 BACK END

PHP

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. Originally created by Rasmus Lerdorf in 1994, the PHP reference implementation is now produced by The PHP Group. PHP originally stood for Personal Home Page, but it now stands for the recursive backronym PHP: Hypertext Preprocessors.

PHP code may be embedded into HTML code. or it can be used in combination with various web template systems. Web content management system and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images. with the generated web page. PHP code may also be executed with a command-line interface (CLI) and can be used to implement standalone graphical applications.

MYSQL

MySQL server is powerful database, and it requires limited programs and used has back end. It supports GUI and more application is developed by help this server. Collection of tables which holds the data is called database. A beginner can create their own database by click home page. ships with no GUI tools to administer MySQL databases or manage data contained within the databases. Users may use the included command line tools or install MySQL Workbench via a separate download. Many third-party GUI tools are also available.

2.4 FEASABILITY ANALYSIS

A feasibility study is an evaluation and analysis of the potential of the proposed project which is based on extensive investigation and research to give full comfort to the decision makers. Feasibility studies aim to objectively and rationally uncover the strength and weakness of existing business of proposed venture, opportunities and threads as presented by the environment, the resources required to carry through, and ultimately the process for success. In its simplest terms, the two criteria to judge feasibility are cost required and value to attain. As such, a well-designed feasibility study should provide a historical background of the business or project, description of the product or service, accounting statements, details of the operations and management, marketing research and policies, financial data, legal requirements and tax obligations.

The four aspects in the feasibility study are:

- Technical feasibility
- Economic feasibility
- Operational feasibility
- Behavioural feasibility

Technical Feasibility

The technical feasibility centres on the existing system and what extend it can support the proposed addition. The technical feasibility assessment is focused on gaining an understanding of the present technical resources of the organization and their applicability to the expected needs of the proposed system. The minimum requirements of the system are met by average user. The developer system has á modest technical requirement as only minimal or null changes are required for implementing system.

Normally associated with the technical feasibility includes:

- Development risk
- Resource availability
- Technology

The proposed system can work without any additional hardware or software support other than the computer system and networks. So, I analysed that the proposed system is much more technically feasible than other systems when comparing with the benefits of the new system.

Economic Feasibility

Economic feasibility analysis is also known as cost/benefit analysis. The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide. The proposed system reduces the operating cost in terms of time by automating the process. This system is economically feasible.

Operational Feasibility

Operational feasibility is a measure of how well a proposed system solves the problems and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development.

Behavioural Feasibility

People are inherently resistant to changes and computer is known for facilitating the changes. An estimate should be made to how strongly the users react towards the development of the system. The proposed system consumes less time. Thus, the people are made to engage in some other important work.

2.5 DATA FLOW DIAGRAM (DFD)

2.5.1 INTRODUCTION TO DATA FLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system. It differs from the flowchart as it shows the data flow instead of the control flow of the program. A data flow diagram can also be used for the visualization of data processing (structured design).

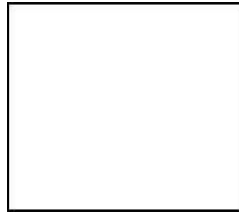
Data flow diagrams were invented by Larry Constantine, the original developer of structured design based on Martin and Estrin's "data flow graph" model of computation.

Data flow diagrams (DFDs) are one of the three essential perspectives of Structured System Analysis and Design Method SSADM. The sponsor of a project and the end users will need to be briefed and consulted throughout all stages of a system's evolution. With a data flow diagram, users can visualize how the system will operate, what the system will accomplish, and how the system will be implemented. The old system's data flow diagrams can be drawn up and compared with the new system's data flow diagrams to draw comparisons to implement a more efficient system. Data flow diagrams can be used to provide the end user with physical idea of where the data they input ultimately has an effect upon the structure of the whole system from order to dispatch to report. How any system is developed can be determined through a data flow diagram.

Developing a data flow diagram helps in identifying the transaction data in the data model. There are different notations to draw data flow diagrams, defining different visual representation for process, data stores, data flow, and external entities. The first step is to draw a data flow diagram (DFD). A DFD also known as "bubble chart" has the purpose of clarifying system requirements and identifying major transformation that will become program in system design. So, it is starting point of the design phase that functionally decompose the requirements specification down to the lowest level of details DFD consists of series of bubbles joined by lines. The bubbles represent data transformation and the lines represent data flow in the system.

DFD Symbols: -

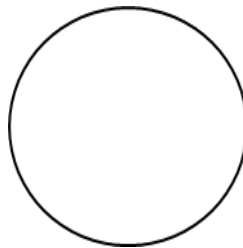
- Square- Defines source or destination of system.



- Data flow - Identifies data flow Circle.

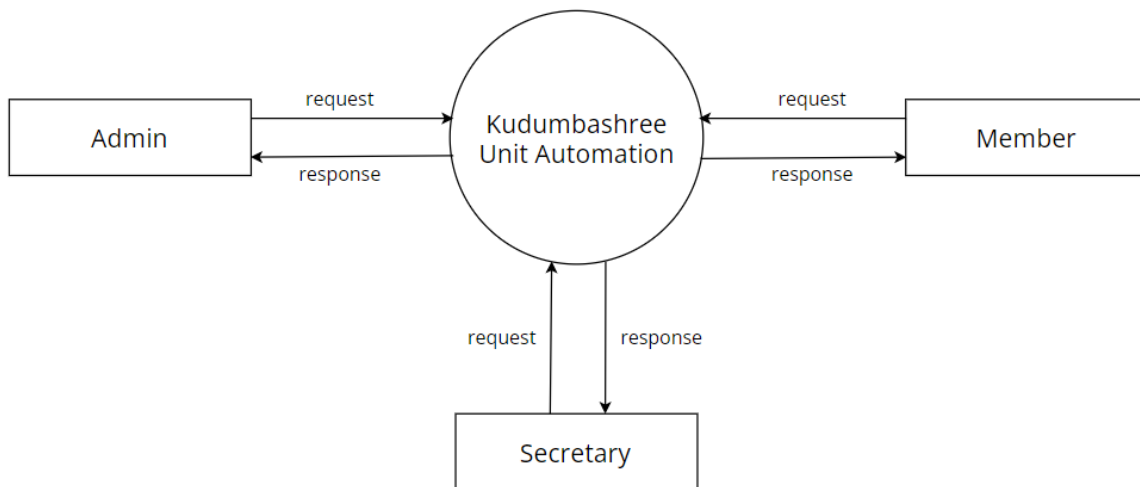


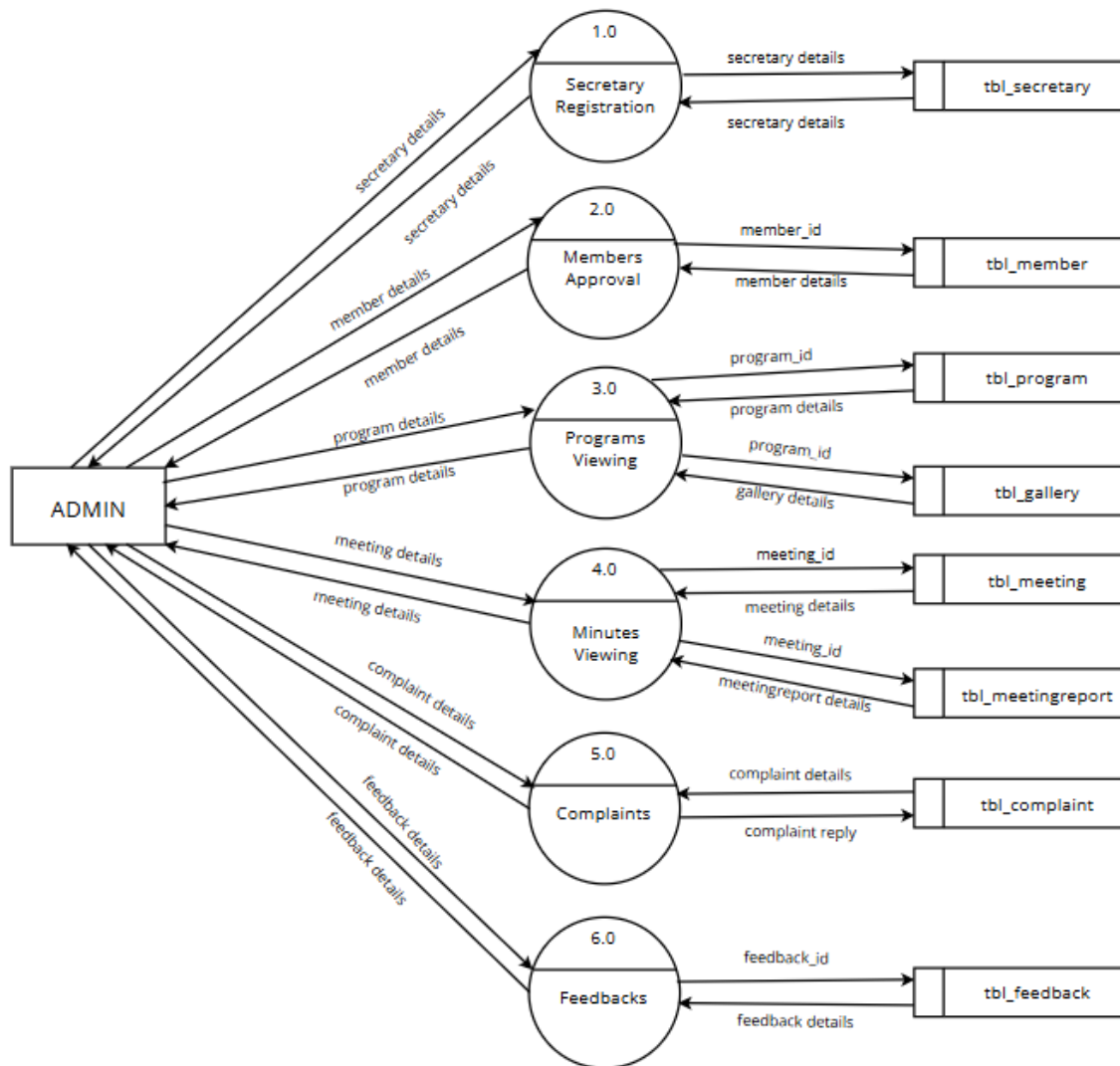
- Bubble - Represents a process that transforms incoming data to outgoing data.

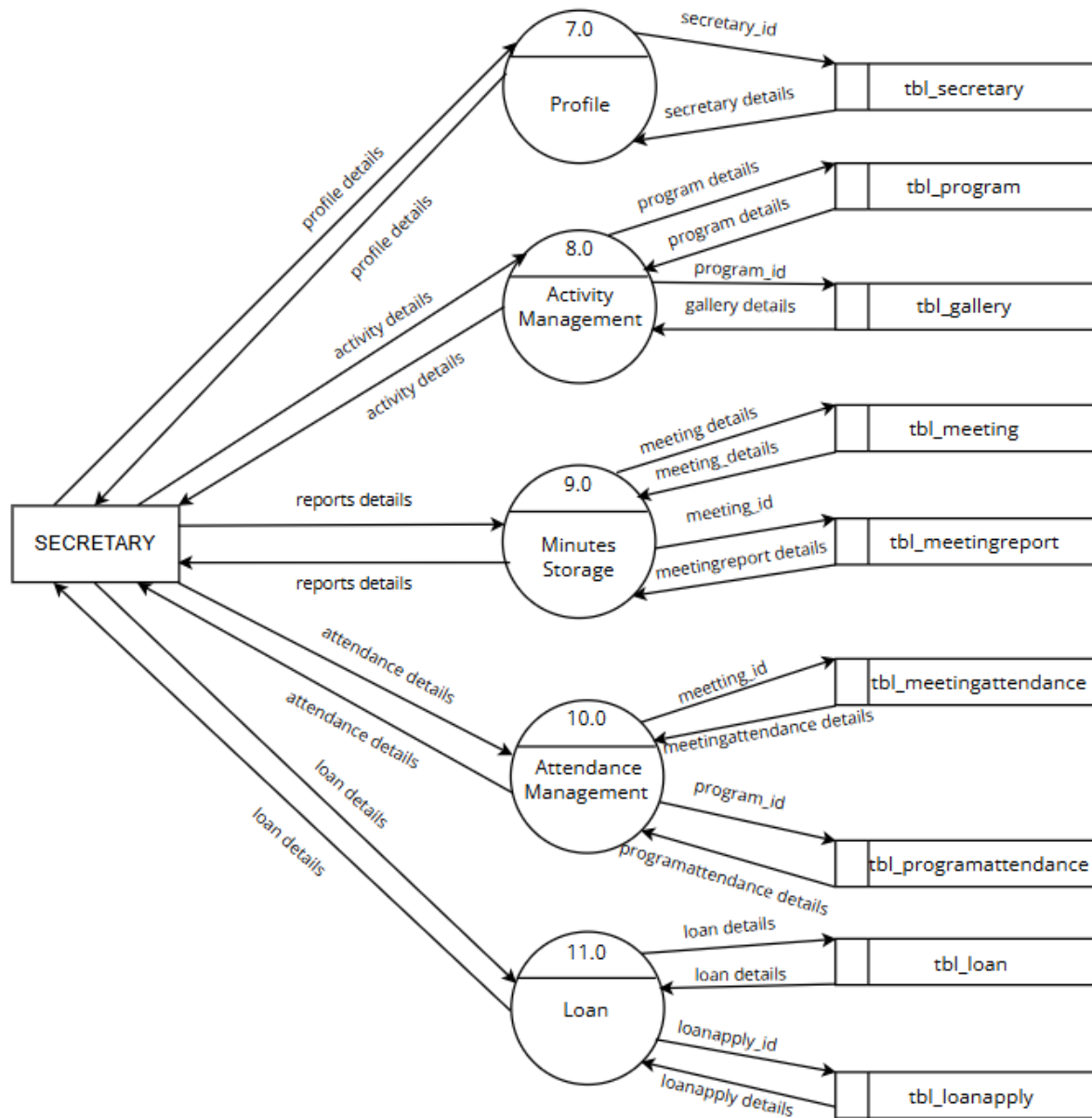


- Open rectangle- Data store



LEVEL-0

LEVEL-1





3. SYSTEM DESIGN

3.1 INPUT DESIGN

The quality of the system input determines the quality of the system output. Input specification describes the way data enter the system for processing. Input design features can ensure the reliability of the system and produce result from accurate data, or they can result in the production of erroneous information. The input design also determines whether the user can interact efficiently with the system.

In our system almost all inputs are being taken from the databases. To provide adequate inputs we have to select necessary values from the databases and arrange it to the appropriate controls.

Admin

Admin is the one who controls the system. The admin can access the page using username and password. Admin can add secretary, approve members and view reports.

Secretary

Secretary is the user of the system. Secretary can add attendance of members. Secretary also adds meeting and program reports. He also shares the details of available loans and users can apply it if needed.

Member

Member is the user of the system. Member can view their attendance, They can also view various reports that secretary added.

3.2 OUTPUT DESIGN

One of the important features of an information system for users is the output produces. Output is the information delivered to users through the information system. Without quality of the output, the entire system appears to be unnecessary that users will avoid using it. Users generally merit the system solely by its output. In order to create the most useful output possible. One works closely with the user through an interactive process. Until the result is considered to be satisfactory.

Admin

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Member

Member is the user of the system. Member can view their attendance, They can also view various reports that secretary added.

3.3 TABLE DESIGN

The data design transforms the information domain model created during analysis into the data structures that will be required to implement the software. The data objects and relationships defined in the entity relationship diagram and the detailed data content depicted in the data dictionary provide the basis for the data design activity.

The overall objective in the development of database technology has been to treat data as an organizational resource and as an integrated whole. Database Management System allows data to be protected and organized separately from other resources. Database is a integrated collection of data. This is the difference between logical and physical data.

The organization of data in the database aims to achieve three major objectives:

- Data integration
- Data integrity
- Data independence

The databases are implemented using a DBMS package. Each DBMS has unique characteristics and general techniques for database design. There are 6 major steps in design process. The first 5 steps are usually done on paper and finally the design is implemented.

- Identify the table and relationships.
- Identify the data that is needed for each table and relationship.
- Resolve the relationship.
- Verify the design.
- Implement the design.

The database uses tables for storage. A table also contains records, which is a set of fields. All records, in a table have the same set of fields with different information uses 3 tables.

Each table contains key fields that establish relationships in the database and how the records are stored. There are primary key fields that uniquely identify a record in a table. There are also fields that contain the primary key from another table called foreign keys.

The various database tables that are used in this project are the following:

1. Table Name: tbl_admin

Primary Key: admin_id

Description: Stores the details of admin

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
admin_id	INT	--	Unique ID of Admin
admin_name	VARCHAR	50	Name of Admin
admin_email	VARCHAR	50	Email of Admin
admin_password	VARCHAR	50	Password of Admin

2. Table name: tbl_secretary

Primary Key: secretary_id

Description: Stores the details of secretary

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
secretary_id	INT	--	Unique ID of Secretary
secretary_name	VARCHAR	50	Name of Secretary
secretary_email	VARCHAR	50	Email of Secretary
secretary_contact	VARCHAR	50	Contact of Secretary
secretary_photo	VARCHAR	100	Photo of Secretary
secretary_proof	VARCHAR	100	Proof of Secretary
secretary_address	VARCHAR	100	Address of Secretary
secretary_password	VARCHAR	50	Password of Secretary

3. Table Name: tbl_member

Primary Key: member_id

Description: Stores the details of members

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
member_id	INT	--	Unique ID of Secretary
member_name	VARCHAR	50	Name of Secretary
member_email	VARCHAR	50	Email of Secretary
member_contact	VARCHAR	50	Contact of Secretary
member_photo	VARCHAR	100	Photo of Secretary
member_proof	VARCHAR	100	Proof of Secretary
member_address	VARCHAR	100	Address of Secretary

member_password	VARCHAR	50	Password of Secretary
member_status	INT	--	Status of Secretary

4. Table Name: tbl_meeting

Primary Key: meeting_id

Description: Stores the details of meetings.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
meeting_id	INT	--	Unique ID of meetings
meeting_date	DATE	--	Date of meetings
meeting_details	VARCHAR	1000	Details of meetings

5. Table Name: tbl_meetingreport

Primary Key: meetingreport_id

Foreign Key: meeting_id

Description: Stores the meeting reports.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
meetingreport_id	INT	--	Unique ID of meeting report
meeting_id	INT	--	Unique ID of meeting
meetingreport_desc	VARCHAR	1000	Description of meeting report

6. Table Name: tbl_program

Primary Key: program_id

Description: Stores the program details.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
program_id	INT	--	Unique ID of program
program_name	VARCHAR	50	Name of program
program_details	VARCHAR	100	Details of program

7. Table Name: tbl_gallery

Primary Key: gallery_id

Foreign Key: program_id

Description: Stores the photos of programs.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
gallery_id	INT	--	Unique ID of photo
gallery_file	VARCHAR	100	Photo of program
program_id	INT	--	Unique ID of program

8. Table Name: tbl_meetingattendance

Primary Key: meetingattendance_id

Foreign Key: member_id, meeting_id

Description: Stores the meeting attendance of members.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
meetingattendance_id	INT	--	Unique ID of meeting attendance
member_id	INT	--	Unique ID of member

meeting_id	INT	--	Unique ID of meeting
meetingattendance_status	INT	--	Status of attendance

9. Table Name: tbl_programattendance

Primary Key: programattendance_id

Foreign Key: member_id, program_id

Description: Stores the program attendance of members.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
programattendance_id	INT	--	Unique ID of meeting attendance
member_id	INT	--	Unique ID of member
program_id	INT	--	Unique ID of program
programattendance_status	INT	--	Status of attendance

10. Table Name: tbl_loan

Primary Key: loan_id

Description: Stores the details of loan.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
loan_id	INT	--	Unique ID of loan
loan_name	VARCHAR	50	Name of Loan
loan_desc	VARCHAR	1000	Description of Loan

11. Table Name: tbl_loanapply

Primary Key: loanapply_id

Foreign Key: loan_id, member_id

Description: Stores the details of loan application.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
loanapply_id	INT	--	Unique ID of loan application
loan_id	INT	--	Name of loan
member_id	INT	--	Name of member
loanapply_status	INT	--	Status of loan application
loanapply_date	DATE	--	Date of loan application
loanapply_content	VARCHAR	1000	Content of loan application

12. Table Name: tbl_complaint

Primary Key: complaint_id

Foreign Key: member_id

Description: Stores the details of complaints.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
complaint_id	INT	--	Unique ID of complaint
complaint_title	VARCHAR	50	Title of complaint
complaint_content	VARCHAR	100	Content of complaint
complaint_reply	VARCHAR	100	Reply of complaint
complaint_date	DATE	--	Date of complaint
member_id	INT	--	Unique ID of member
complaint_status	INT	--	Status of complaint

13. Table Name: tbl_feedback

Primary Key: feedback_id

Foreign Key: member_id

Description: Stores the details of feedbacks.

FIELD NAME	DATATYPE	SIZE	DESCRIPTION
feedback_id	INT	--	Unique ID of feedback
feedback_content	VARCHAR	100	Content of feedback
member_id	INT	--	Unique ID of member

4. SYSTEM TESTING AND IMPLEMENTATION

4.1. SYSTEM TESTING

Testing is the process of examining the software to compare the actual behaviour with that of the expected behaviour. The major goal of software testing is to demonstrate that faults are not present. In order to achieve this goal, the tester executes the program with the intent of finding errors. Though testing cannot show absence of errors but by not showing their presence it is considered that these are not present.

System testing is defined as the process by which one detects the defects in the software. Any software development organization or team has to perform several processes. Software testing is one among them. It is the final opportunity of any programmer to detect and rectify any defects that may have appeared during the software development stage. Testing is a process of testing a program with the explicit intention of finding errors that makes the program fail. In short system testing and quality assurance is a review in software products and related documentation for completion, correctness, reliability and maintainability.

System testing is the first stage of implementation, which is aimed at ensuring that the system works accurately and efficiently before live operation commences. Testing is vital to the success of the system. System testing makes a logical assumption that if all the parts of the system are correct and the goal will be successfully achieved. A series of testing are performed for the proposed system before the proposed system is ready for user acceptance testing.

The testing steps are,

- Unit testing
- Integration testing
- Acceptance Testing
- Validation

- Output testing

System Testing provides the file assurance that software once validated mast combined with all other system elements. System testing verifies whether all elements have been combined properly and that overall system function and performance is achieved. FA the integration of

1. Unit Testing

Unit testing is caried out screen-wise, each screen being identified as an object. Attention is diverted to individual modules, independently to one another to locate errors. This has enabled the detection of errors in coding and logic.

Various test cases are prepared. For each module these test cases are implemented, and it is checked whether the module is executed as per the requirements and outputs the desired result.

In this test each service input and output parameters are checked.

In unit testing

- Module interface was tested to ensure that information properly flows into and out of the program under test.
- Boundary condition was tested to ensure that module operates properly at boundaries established to limit or restrict processing.
- All independent paths through the control structures were executed to ensure that all statements in the modules have been executed at least once.
- Error handling paths were also tested.

2. Integration Testing

Integration testing is a systematic technique for constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing.

Unit tested module were taken and a single program structure was built that has been dictated by the design. Incremental integration has been adopted here.

The modules are tested separately for accuracy and modules are integrated too. Then, using bottom-up integration i.e., by integrating from moving from bottom to the top of the system is checked and errors found during integration are rectified. In this testing individual modules were combined and the module wise Shifting was verified to be alright.

The entire software was developed and tested in small segments, where errors were easy to locate and rectify. Program builds (group of modules) were constructed corresponding to the successful testing of user interaction, data manipulation analysis, and display processing and database management.

3. Validation Testing

Validation testing is done to ensure complete assembly of the error-free software. Validation can be termed successful only if it functions in manner. Reasonably expected by the student under validation is alpha and beta testing. The student-side validation is done in this testing phase. It is checked whether the data passed to each student is valid or not. Entering incorrect values does the validation testing and it is checked whether the errors are being considered. Incorrect values are to be discarded. The errors are rectified.

In “Kudumbashree Unit Automation” verifications are done correctly. So, there is no chance for users to enter incorrect values. It will give error messages by using different validations. The validation testing is done very clearly and found it is error free.

4. Output Testing

After performing the validation testing the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in a specific format.

The output format on the screen was found to be correct as the format was designed in the system design phase according to the user needs. For the hard copy also, the output comes out as specified requirement by the user. Hence output testing does not result in any Correction in the system.

This project is developed based on the user choice. It is user friendly. The output format is very clear to user. Output testing is done on Kudumbashree Unit Automation correctly.

5. Acceptance Testing

Acceptance involves running a suite of tests on the completed system. Each individual test, known as a Case, exercise particular operating condition of the operating condition of the user's environment or feature of the system, and will result in a pass fail, or Boolean outcome.

4.2 SYSTEM IMPLEMENTATION

The implementation is the final state, and it is an important phase. It involves the invalid programming system testing. user training and the operational running of developed proposed system that constitutes the application subsystems. A major task of preparing for implementation is education of users. which should really have been taken place much carrier in the project when they were belong involved in the investigation and design work. During the implementation phase system take physical shape. In order to develop a system implemented planning is very essential.

The implementation phase of the software development is concerned with translating design specification into source code. The user tests the developed system and changes are made according to their needs. Our system has been successfully implemented.

Before implementation several tests have been conducted to ensure that no errors are encountered during the operation. The implementation phase ends with an evaluation of the system after placing into the operation for a period of time.

The process of putting the developed system in actual use is called system implementation. This includes all those activities that take place to convert from old system to new system. The system

can be implemented only after testing is done and is found to be working to specifications. The implementation stage is a systems project in its own right.

The implementation stage involves following tasks:

- Careful planning.
- Investigation of system and constraints.
- Design of method to achieve change over.
- Evaluation of the changeover method.

In the case of this project all the screens are designed first. For making it to be executable, codes are written on each screen and performs the implementation by creating the database and connecting to the server. After that the system, is Checked, whether it performs all the transactions correctly. Then databases are cleared and made it to be usable to the technicians

Implementation Plans

The following are the step involved in the implementation plan of "Kudumbashree Unit Automation":

- Test system with sample data
- Detection and correction of errors
- Make the necessary changes in the system.
- Check the existing system.
- Installation of hardware and software utilities
- Training and involvement of user personals

5.SECURITY TECHNOLOGIES & POLICIES

The protection of computer-based resources that includes hardware, software, data procedures and people against unauthorized use or natural disaster is known as System Security.

System Security can be divided into four related issues:

- Security
- Integrity
- Privacy
- Confidentiality

SYSTEM SECURITY refers to the technical innovations and procedures applied to the hardware and operation systems to protect against deliberate or accidental damage from a defined threat.

DATA SECURITY is the protection of data from loss, disclosure, modification and destruction.

SYSTEM INTEGRITY refers to the proper functioning of hardware and programs, appropriate physical security and safety against external threats such as eavesdropping and wiretapping.

PRIVACY defines the rights of the user or organizations to determine what information they are willing to share with or accept from others and how the organization can be protected against unwelcome, unfair or excessive dissemination of information about it.

CONFIDENTIALITY is a special status given to sensitive information in a database to minimize the possible invasion of privacy. It is an attribute of information that characterizes its need for protection.

SECURITY IN SOFTWARE System security refers to various validations on data in form of checks and controls to avoid the system from failing. It is always important to ensure that only valid data is entered, and only valid operations are performed on the system.

The system employees two types check and controls:

CLIENT-SIDE VALIDATION Various client-side validations are used to ensure on the client side that only valid data is entered. Client-side validation saves server time and load to handle invalid data. Some checks imposed are:

- Forms cannot be submitted without filling up the mandatory data so that manual mistakes of submitting empty fields that are mandatory can be sorted out at the client side to save the server time and load.
- Tab-indexes are set according to the need and taking into account the ease of user while working with the system.

SERVER-SIDE VALIDATION Some checks cannot be applied at client side. Server-side checks are necessary to save the system from failing and intimating the user that some invalid operation has been performed or the performed operation is restricted. Some of the server-side checks imposed is:

- Server-side constraint has been imposed to check for the validity of primary key and foreign key. A primary key value cannot be duplicated. Any attempt to duplicate the primary value results into a message intimating the user about those values through the forms using foreign key can be updated only of the existing foreign key values.
- User is intimating through appropriate messages about the successful operations or exceptions occurring at server side.
- Various Access Control Mechanisms have been built so that one user may not agitate upon another. Access permissions to various types of users are controlled according to the organizational structure. Only permitted users can log on to the system and can have access

according to their category. User- name, passwords and permissions are controlled o the server side.

- Using server-side validation, constraints on several restricted operations are imposed.

6. MAINTENANCE

Software maintenance is the modification of a software product and delivery to correct faults, to improve performance or other attributes. Maintenance is the ease with which a program can be corrected if any error is encountered, adapted if its environment changes or enhanced if the customer desires a change in requirement. Maintenance follows conversation to extend that changes are necessary to maintain satisfactory operations relative to changes in the user's environment.

Maintenance often includes minor enhancements or corrections to problems that surface in the system's operation. Maintenance is also done based on fixing the problems reported, changing the interface with other software or hardware enhancing the software.

CATEGORIES OF MAINTENANCE

Corrective Maintenance

Corrective maintenance is the most used maintenance approach, but it is easy to see its limitations. When equipment fails, it often leads to downtime in production, and sometimes damages other parts. In most cases, this is expensive. Also, if the equipment needs to be replaced, the cost of replacing it alone can be substantial. Reliability of systems maintained by this type of maintenance is unknown and cannot be measured. Corrective maintenance is possible since the consequences of failure or wearing out are not significant and the cost of this maintenance is not great.

Perfective Maintenance

Modification of a software product alter delivery to improve performance or maintainability. This term is used to describe changes undertaken to expand the existing requirements of the system. A successful piece of software tends to be subjected to the Succession of changes resulting in an increase in its requirements. This is based on premise that as the software becomes useful, the user experiment with new cases beyond the of Scope for which it was initially developed. The requirements can take the enhancement form of existing system functionality and improvement in computational efficiency.

Adaptive Maintenance

Modification of a software product performed after delivery to keep a product usable, changed or changing environment. Adaptive maintenance includes any work initiated because of moving the software to a different hardware or software platform. It is a change driven by the need to accommodate modifications in the environment of software system. The environment in this context refers to the totality of all conditions and influences which act from outside upon the system. A change to the whole or part of this environment will Warrant a corresponding modification of the software.

Preventive Maintenance

Preventive maintenance is a schedule of planned maintenance actions aimed at the prevention of breakdowns and failures. The primary goal of preventive maintenance is to prevent the failure of equipment before it occurs. It is designed to preserve and enhance equipment reliability by replacing worn components before they fail. Preventive maintenance activities include equipment checks, partial or complete overhauls at specified periods.

Long-term benefits of preventive maintenance include:

- Improved system reliability.
- Decreased cost of replacement.
- Decreased system downtime.

7. SCOPE FOR FUTURE ENHANCEMENT

The **Kudumbashree Unit Automation** project offers several avenues for future enhancements to further improve its functionality and reach. Key areas for development include integrating the system with government portals for real-time data exchange, developing a mobile application for increased accessibility, and incorporating advanced analytics for better decision-making. Additionally, implementing multilingual support would ensure inclusivity, while e-payment integration could streamline financial transactions. Introducing AI-based task automation and real-time communication tools would enhance efficiency and collaboration. Future upgrades could also focus on scalability to accommodate larger units, along with strengthening data security to protect sensitive information. These enhancements would not only improve the system's usability but also ensure its sustainability and scalability for future needs.

8. CONCLUSION

The **Kudumbashree Unit Automation** project aims to enhance the operational efficiency and digital transformation of Kudumbashree units. By automating key processes such as member management, financial tracking, and activity scheduling, this system reduces manual effort, minimizes errors, and ensures smoother coordination.

This project showcases the importance of integrating technology into community-based initiatives, offering a scalable and user-friendly solution that can be expanded to other similar units in the future. The automation system not only aligns with the goals of empowering local communities but also serves as a step towards a more sustainable and digitally empowered society.

Through this project, we have gained practical insights into software development, database management, and the impact of digital tools in grassroots-level initiatives, setting the foundation for further enhancements and scalability.

9. BIBLIOGRAPHY

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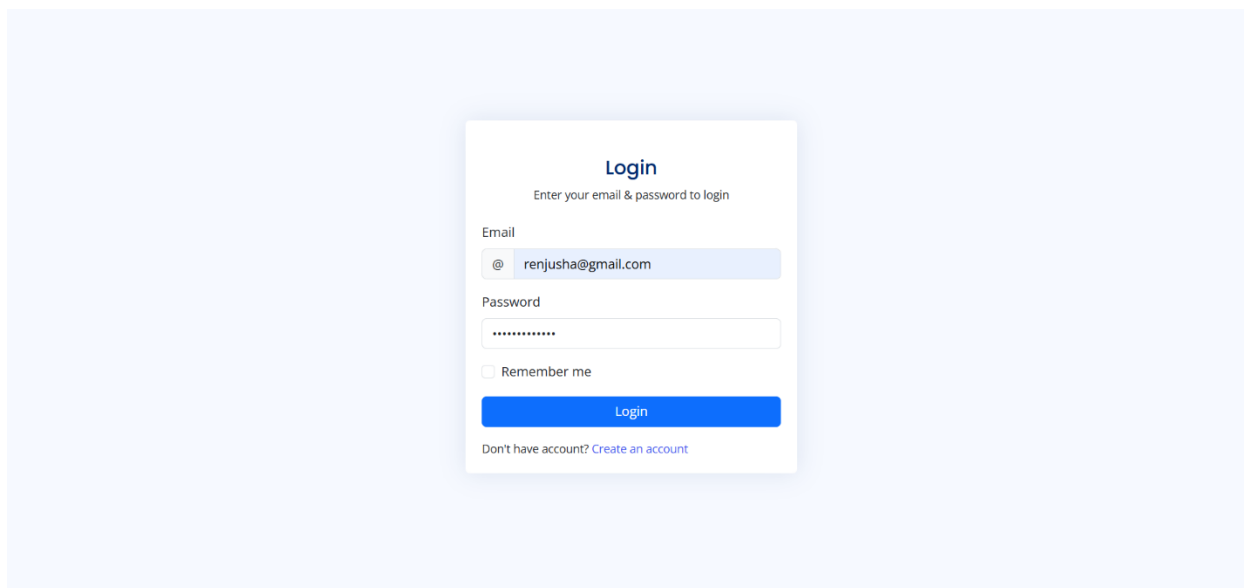
10. APPENDIX

10.1 SCREENSHOTS

Homepage



Login



Sign Up

Sign Up

Enter your personal details to create account

Name:

Contact:

Nisha Satheesan

9744034945

Address:

Parachalil (H), Kombanad. P.O

Photo:

Choose File

No file chosen

Proof:

Choose File

No file chosen

Email

Password

nisha@gmail.com

Submit

Reset

[Back to Login](#)

Admin Homepage

Kudumbashree.All

Home

Add Secretary

Members Approval

Programs

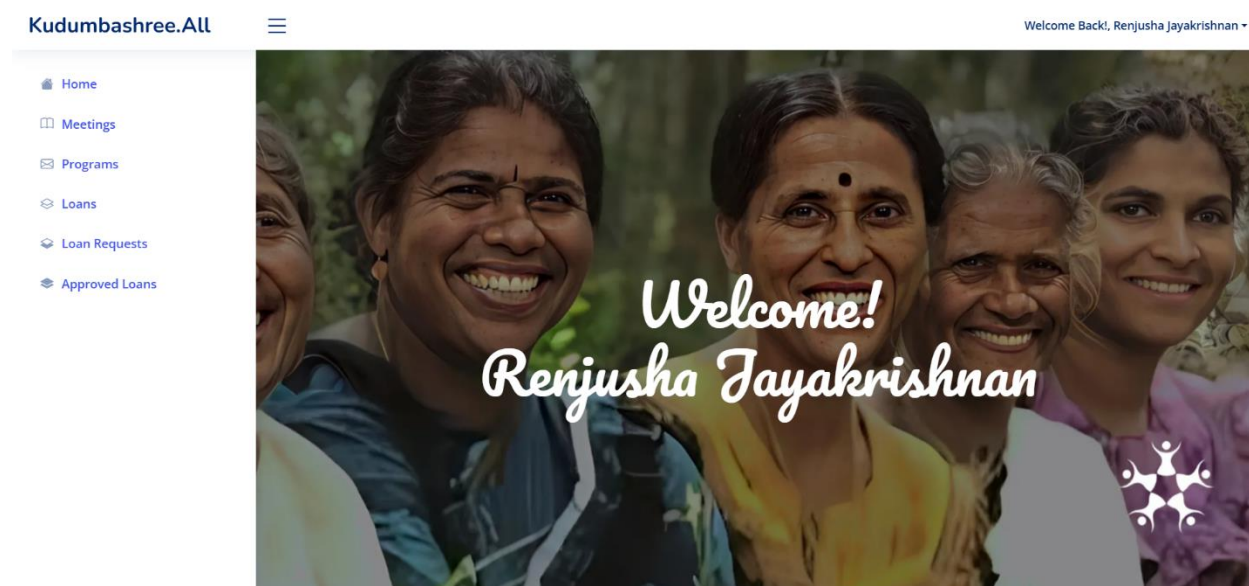
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Secretary Homepage



Member Homepage



Add Secretary

Kudumbashree.All
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Add Secretary

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Enter Details:

Name:

Address:

Contact:

Photo:

Proof:

Email:

Password:

Members Approval

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Members Approval

Home / Members Approval

Pending

Sl No.	Name	Address	Contact	Photo	Proof	Email	Password	
1	Omana Anirudhan	Pulimoottikudy (H), Kombanad.P.O	7894561230			omana@gmail.com	Omana@2024	Accept Reject

Accepted

Sl No.	Name	Address	Contact	Photo	Proof	Email	Password	Status
1	Saradha Narayanan	Njarlakkattukudy (H), Kombanad.P.O	9745739028			saradha@gmail.com	Saradha@2024	Accepted Reject
2	Devaki Velayudhan	Parachalil (H), Kombanad.P.O	7034226825			devaki@gmail.com	Devaki@2024	Accepted Reject

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20th Varshikam

Family reunion and Kudumbashree Varshikam was held on 20-10-2024 at residence of Shailaja Sudhakaran. The meeting started by lighting the nilavilaku. The members sang the prayer together. Renjusha Jayakrishnan delivered the welcome speech. The unit was reminded to improve its functioning and to fill the shortcomings. Then Renjusha Jayakrishnan presented the report. There was no significant discussions about the report. 2 folk songs were sung by the unit members. Ex-secretary Saradha Narayanan and current ADS member Rejitha Sajeew delivered the greeting speech. All members sung the National Anthem.

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2024-10-27

The 1019th meeting of Pulary Kudumbashree Unit was held at the residence of Shailaja Sudhakaran. Presidential speech was conveyed by Kanaka Gangadharan. Suprabha Murali conveyed the welcome speech.

The meeting started at 2pm with a prayer. The ADS meeting of 13th ward is scheduled on 28th October 2024. All the members is requested to participate in the meeting. The meeting takes place at Govt. U. P School Kombanad. The collection amount is Rs.100. Thrift - 250, Loan - 200, Annual collection - 100. Rs. 550 was given to Renjusha jayakrishnan.

Profile

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Welcome Back!, Renjusha Jayakrishnan



Renjusha Jayakrishnan
Secretary

Overview Edit Profile Change Password

Profile Details

Full Name	Renjusha Jayakrishnan
Address	Madathedam (H), Kombanad
Role	Secretary
Contact	7894561230
Email	renjusha@gmail.com
Password	Renjusha@2024

Meeting

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Details

The 1019th meeting of Pulary Kudumbashree Unit was held at the residence of Shailaja Sudhakaran. Presidential speech was conveyed by Kanaka Gangadharan. Suprabha Murali conveyed the welcome speech.

Submit

Meetings

Date	Details			
2024-10-27	The 1019th meeting of Pulary Kudumbashree Unit was held at the residence of Shailaja Sudhakaran. Presidential speech was conveyed by Kanaka Gangadharan. Suprabha Murali conveyed the welcome speech.	Delete	Add Attendance	Minutes

Minutes Report

Department of Computer Science, Nirmala College Muvattupuzha

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Select Member

Member
--SELECT--

Present
Absent

Program

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Name

20th Varshikam

Details

Family reunion and Kudumbashree Varshikam was held on 20-10-2024 at residence of Shailaja Sudhakaran. The meeting started by lighting the nilavilaku. The members sang the prayer together. Renjusha Jayakrishnan delivered the welcome speech. The unit was reminded to improve its functioning and to fill the shortcomings. Then Renjusha Jayakrishnan presented the report. There was no significant discussions about the report. 2 folk songs were sung by

Submit
Reset

Programs

Name	Details	
20th Varshikam	Family reunion and Kudumbashree Varshikam was held on 20-10-2024 at residence of Shailaja Sudhakaran. The meeting started by lighting the nilavilaku. The members sang the prayer together. Renjusha Jayakrishnan delivered the welcome speech. The unit was reminded to improve its functioning and to fill the shortcomings. Then Renjusha Jayakrishnan presented the report. There was no significant discussions about the report. 2 folk songs were sung by the unit members. Ex-secretary Saradha Narayanan and current ADS member Reliitha Saieev delivered the greeting speech. All members sung the	Delete Add Attendance Add Photo

Loan

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Welcome Back!, Renjusha Jayakrishnan

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Enter Loan Details

Name
Loan 1

Details

Dear Members, We are pleased to inform you that our Kudumbashree Unit is now accepting applications for loan 1 to support our members financial needs. This loan facility aims to provide financial assistance to members for personal and family purposes. Please review the details below for eligibility and application guidelines: Loan Details: (Loan Amount: Up to Rs.8000 Interest Rate: 2% per monthly Repayment Period: 6 months).

Submit
Reset

Loans

Name	Description	
Loan 1	Dear Members, We are pleased to inform you that our Kudumbashree Unit is now accepting applications for loan 1 to support our members financial needs. This loan facility aims to provide financial assistance to members for personal and family purposes. Please review the details below for eligibility and application guidelines: Loan Details: (Loan Amount: Up to Rs.8000 Interest Rate: 2% per monthly Repayment Period: 6 months).	Delete

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Welcome Back!, Renjusha Jayakrishnan

Loan Requests
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Member Name	Contact	Loan	Content	
Saradha Narayanan	9745739028	Loan 1	I am Saradha narayanan, a member of Pulari Kudumbashree Unit, and I am writing to apply for a loan to support my sons education. I am requesting a loan of ₹8000 and propose to repay it over 6 months. This financial support would be immensely helpful in my sons edcational purpose. I assure you that I will adhere to the repayment terms as set by the unit. Please let me know if any further information or documents are needed for this application. Thank you for considering my request. I look forward to a positive response.	Approve Reject

Approved Loan

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Welcome Back!, Renjusha Jayakrishnan

Approved Loans

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Member Name	Contact	Loan	Content
Saradha Narayanan	9745739028	Loan 1	Dear Secretary, I am Saradha Narayanan, a member of Pulari Kudumbashree, and I am writing to apply for this loan for my sons educational purpose. I propose to repay it over 6 months. This financial support would be immensely helpful for my sons education. I assure you that I will adhere to the repayment terms as set by the unit. Please let me know if any further information or documents are needed for this application. Thank you for considering my request. I look forward to a positive response.

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Give Feedback

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Date	Status
2024-10-27	Present

Apply Loan

Kudumbashree.All

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My Loan
Give Complaint
Give Feedback

Welcome Back!, Saradha Narayanan

Apply Loan

Home / Loans / Apply Loan

Enter request

Content

I am Saradha narayanan, a member of Pulari Kudumbashree Unit, and I am writing to apply for a loan to support my sons education. I am requesting a loan of ₹8000 and propose to repay it over 6 months. This financial support would be immensely helpful in my sons educational purpose. I assure you that I will adhere to the repayment terms as set by the unit. Please let me know if any further information or documents are needed for this application. Thank you for

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Reset

Loan Status

Kudumbashree.All

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Give Complaint
Give Feedback

Welcome Back!, Saradha Narayanan

My Loan

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Your Loan Status

Loan Name	Description	Content	Date	Status
Loan 1	Dear Members, We are pleased to inform you that our Kudumbashree Unit is now accepting applications for loan 1 to support our members financial needs. This loan facility aims to provide financial assistance to members for personal and family purposes. Please review the details below for eligibility and application guidelines: Loan Details: (Loan Amount: Up to Rs.8000 Interest Rate: 2% per monthly Repayment Period: 6 months).	Dear Secretary, I am Saradha Narayanan, a member of Pulari Kudumbashree, and I am writing to apply for this loan for my sons educational purpose. I propose to repay it over 6 months. This financial support would be immensely helpful for my sons education. I assure you that I will adhere to the repayment terms as set by the unit. Please let me know if any further information or documents are needed for this application. Thank you for considering my request. I look forward to a positive response.	2024-10-28	Approved
Loan 1	Dear Members, We are pleased to inform you that our Kudumbashree Unit is now accepting applications for loan 1 to support our members financial needs. This loan facility aims to provide financial assistance to members for personal and family purposes. Please review the details below for eligibility and application guidelines: Loan Details: (Loan Amount: Up to Rs.8000 Interest Rate: 2% per monthly Repayment Period: 6 months).	I am Saradha narayanan, a member of Pulari Kudumbashree Unit, and I am writing to apply for a loan to support my sons education. I am requesting a loan of ₹8000 and propose to repay it over 6 months. This financial support would be immensely helpful in my sons educational purpose. I assure you that I will adhere to the repayment terms as set by the unit. Please let me know if any further information or documents are needed for this application. Thank you for considering my request. I look forward to a positive response.	2024-10-30	Pending